

PAPER_1090: Dark Energy Buoyancy Sector Lagrangian with Euler-Lagrange Residual Analysis

Abstract

We derive the Lagrangian for the dark energy buoyancy sector:

$$\mathcal{L}_{\text{DE}} = \rho_{\text{SCm}} \cdot c^2 \cdot S_{26} \cdot \Phi \cdot (2R - 1) \cdot V$$

where $V = 10^{48} \text{ m}^3$ is an effective cosmological volume, $R = F_{U,Bi}/F_U$ is the buoyancy ratio, and $\Phi = \Phi_0 \cdot S_{26}$ is the phonon amplitude. The Euler-Lagrange equation $\delta S/\delta \phi_{\text{DE}} = 0$ is solved and the residual characterizes three distinct cosmological regimes: accelerating ($R > 0.5$), decelerating ($R < 0.5$), and balanced ($R = 0.5$).

§1 Lagrangian Density

$$\mathcal{L}_{\text{DE}} = \rho_{\text{SCm}} \cdot c^2 \cdot S_{26} \cdot \Phi \cdot (2R - 1) \cdot V$$

Symbol	Value	Meaning
ρ_{SCm}	$9.47 \times 10^{-27} \text{ kg/m}^3$	SCm vacuum density
c	$2.998 \times 10^8 \text{ m/s}$	Speed of light
S_{26}	1.8594...	26-layer Ramanujan sum
Φ_0	10^{20}	Phonon base amplitude
R	$F_{U,Bi}/F_U$	Buoyancy ratio
V	10^{48} m^3	Effective volume

§2 Euler-Lagrange Equation

Varying the action $S = \int \mathcal{L}_{\text{DE}} dt$ with respect to the dark energy field ϕ_{DE} :

$$\frac{\partial \mathcal{L}}{\partial \phi_{\text{DE}}} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_{\text{DE}}} = 0$$

The EL residual is:

$$\mathcal{R}_{\text{EL}} = \rho_{\text{SCm}} c^2 S_{26} \Phi V \cdot \frac{\partial}{\partial \phi_{\text{DE}}} (2R - 1) - \frac{d}{dt} \left[\rho_{\text{SCm}} c^2 S_{26} V \cdot \frac{\partial (\Phi (2R - 1))}{\partial \dot{\phi}_{\text{DE}}} \right]$$

At stationarity, $\mathcal{R}_{\text{EL}} = 0$ exactly.

§3 Three Cosmological Regimes

The sign of $(2R - 1)$ partitions the dynamics:

Regime	Condition	$\mathcal{L}_{\text{DE}}$	Cosmology
Accelerating	$R > 0.5$	> 0	Positive dark energy, cosmic acceleration
Balanced	$R = 0.5$	$= 0$	Zero dark energy, Milne-like coasting
Decelerating	$R < 0.5$	< 0	Negative dark energy, cosmic deceleration

§4 Benchmark Values

At the solar benchmark ($M = M_{\odot}$, $r = 1$ AU):

$$\mathcal{L}_{\text{DE}} = 9.47 \times 10^{-27} \times (2.998 \times 10^8)^2 \times 1.86 \times 1.86 \times 10^{20} \times (2 \times 0.8 - 1) \times 10^{48}$$

$$\mathcal{L}_{\text{DE}} \approx 1.77 \times 10^{47} \text{ J}$$

with EL residual $\mathcal{R}_{\text{EL}} \sim 10^{-12}$ (near-stationarity at benchmark).

§5 Connection to w_{DE}

The Lagrangian generates the equation of state (PAPER_1087):

$$w_{\text{DE}} = \frac{p_{\text{DE}}}{\rho_{\text{DE}} c^2} = -1 + \frac{\partial \ln \mathcal{L}_{\text{DE}}}{\partial \ln a}$$

where $a(t)$ is the scale factor from PAPER_1084.

References

- PAPER_1088: $F_{U,Bi,i}$ Seven-Component Decomposition
- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1086: SCm Dark Energy Density with Γ -Coupling
- PAPER_1087: Dark Energy Equation of State w_{DE}